Tamagachi

Section 1: Project Overview

I am making a physical tamagachi which has two input buttons, pet and feed, to interact and take care of the pet inside.

The Tamagachi will display an ASCII screen containing the pet's face and additional UI. The user would interact with this screen using the two buttons that switches between pet and feed or move and select. In addition, there is a RGB LED light that represents the mood of the pet.

In this project, I will design and create a physical circuit consisting of an LCD, LED, 2 buttons, a breadboard, an Arduino, and wires. This circuit would be controlled/activated by the Arduino which handles hardware logic(what is displayed, what components are activated and how so) based on events such as the user clicking the buttons. Then, it will all be encased in a 3D printed case which I will design and model using CAD.

Principles:

- DC (direct current) - Electricity flows in one direction
    - I am using a battery which uses direct current, so it is important to use parts that are suited for DC current, like LEDs.
- Ohm's Law (V = IR) - Defines the relationship between voltage, current, and resistance.
    - Using this law prevents me from burning my LED or other parts with excess voltage or current.
- Parallel and Series Circuits
    - Parallel - Voltage is equal across all branches and current is split between branches depending on the resistance. Total resistance is the sum of the reciprocals of each branch' resistance.

    - *Series - Voltage changes throughout the circuit with resistors and current is constant. Voltage is found by adding the voltage drops, where the sum of all of them equals the supply voltage.*
        - *This helps me design my circuit so then the LED has the correct brightness and my components don't burn out.*

My project will function via user button inputs, feed and play, which will send different signals to the software. The software will interpret those into logic and activate the hardware for the correct response for the user to see.

Success

- *LCD shows different facial expressions based on pets' mood*
- *Pet has 3 Moods, happy, neutral, and sad*
- *Has feed button that updates the hunger value*
- *Mood changes based on hunger*
- *Has internal clock that increases hunger as time goes on*

Section 2: Engineering or Experiment Breakdown

Phase 1 – Research and Design

Finalize materials

- *Tinker with materials already obtained to confirm functionality*
- *Add missing materials to materials list*
- *Update materials list with Amazon links*

Research

- *Learn basic controls of Arduino*
- *Learn basic Arduino c++ programming*
- *Learn basic 3D printing design principles*

Design

- *Research common embedded systems structure*
- *Plan software-hardware data pipeline*
- *Identify required features and nice-to-have features*
- *Plan and sketch UI and user actions with flowcharts*
- *Record parts dimensions with accompanying sketches*
- *Sketch case design*
- *Sketch circuit schematic*
- *Update materials list if needed*
- *Separate the project into stages and make checkpoints*

Safety

- *Make sure no parts are damaged*
- *Research part voltage/current specifications*
- *Check for potential short circuits and current/voltage overload before connecting battery*

Phase 2 – Build, Develop, or Conduct

1. *Gather electronic components and parts needed*
2. *Display text on LCD*
3. *Connect buttons to edit the LCD screen*
4. *Program state update functions and internal logic*
5. *Link buttons to internal logic*

6. Display facial expression based on the state data
7. Disconnect everything and display different colors with LED
8. Add everything back
9. Program LED to blink to button presses and change color based on mood
10. Finish internal code
11. Sketch and CAD the case
12. 3D print the case and test the fits, iterate on dimensions if needed
13. Test build
14. Polish with more internal logic if time

List step-by-step actions:

- Material preparation
- Construction, programming, or setup
- Data collection (if applicable)
- Adjustments and troubleshooting

Phase 3 – Testing and Iteration

I will do a full run of functionality, testing all inputs and comparing the visual outputs to the internal data. I will also let other people test it, potentially finding more bugs and improvements that could be made to the UI, then I will adjust the program accordingly. I will repeat this process of testing and adjusting until it is complete.

If all of the main functionality is working, I will add more cosmetic details to the case, potentially adding a carabiner, button covers, and extra details. In addition, I would add more software functionality, allowing the user to choose the pet's name, text blinking animations, idle animations, wakeup-sleep timeout, and a buzzer for sound. If there is a lot of time, I can add multiple pet capabilities.

Section 3: Weekly Milestones

Use the following calendar:

For each week, identify:

Week 1: 3/23–3/27

- *Fill in information in planning doc for polishing later*
- *Gather materials needed*
- *Finalize materials list*
- *Finalize features*

Week 2: 3/30–4/3

- *Finalize and submit this planning doc*
- *Display text on LCD*
- *Create pet state structure*
- *Program buttons to update happiness and hunger*
- *Create conditionals to assign mood*
- *Display the happiness and hunger variables after button inputs*
- *Make a timer to make happiness and hunger go down over time*

Week 3: 4/6–4/10

- *Make ASCII facial expressions display based on mood*
- *Make LED light with button presses*
- *Connect LED color to pets' mood*
- *Make pet interactions based on the time since last interaction*

Week 4: 4/13–4/17

- *Connect facial expression to the change of happiness and hungry values*
- *Measure the dimensions of every part*
- *Sketch the case design*

- Start 3D modelling the case

Week 5: 4/20–4/24

- Finalize circuit design and electronics hardware
- Finish CAD of the case
- Print and iterate on case dimensions until assembly works
- Do final print and assemble with any needed soldering or sanding
- Start polishing code and testing features
- If code is polished, start adding polish features

Week 6: 4/27–5/1

- Add polish features in the software
- Update documentation

Final Presentation: 5/9

- Presentation slides with pictures of the build
- Graphs showing user action interactions and data pipeline
- Bullet points of features and information
- Demonstration of the build

Presentation Checkpoint

- Picture of pet state structure - Week 2
- Picture of final electronic circuit - Week 5
- Picture of initial CAD design and final model - Week 5
- Plan presentation structure - Final Week
- Create canva slides and choose template - Final Week
- Inject pictures and information into slides - Final Week
- Format and finalize slides - Final Week
- Practice presenting - Final Week

Section 4: Risk and Contingency Plan

If the project is incompletable in the time frame, I can switch to my software project, the Minesweeper app which is faster, but I would at least put in a week of building before switching.

If the development time exceeds the amount of time I have, I will cut the LED light and just display the mood in the LCD and can cut the CAD case and put the parts in a cardboard box or present with the raw circuit. Lastly, I can simplify the moods and only have one input for interaction.

I can't guarantee I can consistently stay on schedule because I have a research project, TSA polishing, club events, and family/life stuff, so my time is controlled by those. Still, I will work on the project throughout the class period and maybe at lunch too to ensure adequate progress.